Lecture notes: The hosts file (Section 2)

Corrected notes for the lecture "The hosts file". Part A is meant to be pasted into the lecture description on Udemy. Part B is the full text, attached as a resource with the ansible-core handout.

Part A: text for the lecture description (paste as is)

> **تصحيح مهم (نوفمبر 2026):** لو بتستخدم ansible-core 2.19 أو أحدث، اكتب اسم المتغير بتاع مفتاح الـ SSH كده:
>
> `ansible_ssh_private_key_file=~/.ssh/id_rsa`
>
> ومتستخدمش `ansible_private_key`. من أول 2.19 المتغير ده بقى معناه **محتوى** المفتاح نفسه مش مسار الملف، فلو كتبت فيه المسار هيظهر لك الخطأ: `Cannot utilize private_key with SSH_AGENT disabled`
>
> وكمان اسم متغير الاتصال هو `ansible_connection=ssh`. ملف hosts كامل ومصحح، وملف "إيه اللي اتغير في ansible-core 2.19 و 2.20"، موجودين في الـ Resources بتاعة المحاضرة دي.

Correction (Nov 2026): on ansible-core 2.19 or newer, set the key path with `ansible_ssh_private_key_file=~/.ssh/id_rsa`. Don't use `ansible_private_key`: since 2.19 that variable holds the key *contents*, not a path, so a path there fails with "Cannot utilize private_key with SSH_AGENT disabled". The connection variable is `ansible_connection=ssh`. A corrected hosts file and the "What changed in ansible-core 2.19 and 2.20" handout are in this lecture's resources.

Part B: full notes

What the hosts (inventory) file does

The inventory tells Ansible **which machines** to manage and **how to connect** to them. In the course it is a plain INI file called `hosts`, and we pass it with `-i hosts`.

Step 1: get the VM's IP

```
multipass list
multipass info ansiblemain      # the IPv4 line
```

Step 2: one host with everything on one line

```
ansiblemain ansible_host=10.173.4.156 ansible_user=ansibleadmin
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Variable	Meaning
`ansiblemain`	The name Ansible uses for this host (any name you like)
`ansible_host`	IP or DNS name to connect to
`ansible_user`	SSH user (created by cloud-init in `ansible_multipass.yaml`)
`ansible_ssh_private_key_file`	**Path** to the private key that matches the public key in cloud-init

Step 3: group the hosts and share the variables

```
ansiblemain ansible_host=10.173.4.156

[my_hosts]
ansiblemain

[my_hosts:vars]
ansible_connection=ssh
ansible_user=ansibleadmin
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Step 4: test

```
ansible -i hosts my_hosts -m ansible.builtin.ping
ansible -i hosts ansiblemain -m ansible.builtin.setup
```

Corrections compared with the video

In the video or old notes	Correct	Why
`ansible_private_key=...`	`ansible_ssh_private_key_file=...`	In 2.19+, `ansible_private_key` means key **contents**. Error: `Cannot utilize private_key with SSH_AGENT disabled`
`ansible_connect=ssh`	`ansible_connection=ssh`	Typo. Ansible ignores unknown variables, so it silently did nothing (ssh is the default anyway)
`host_key_checking=False` in the inventory	`host_key_checking = False` under `[defaults]` in `ansible.cfg`	It's a config setting, not an inventory variable
`ansible_distribution` in playbooks	`ansible_facts['distribution']`	Injected facts are deprecated in 2.20

Common errors

Error	Cause	Fix
`Cannot utilize private_key with SSH_AGENT disabled`	Path put in `ansible_private_key`	Rename to `ansible_ssh_private_key_file`
`Permission denied (publickey)`	Wrong key, or wrong `ansible_user`	Check the public key in `ansible_multipass.yaml` matches the private key path
`Host key verification failed`	VM recreated with the same IP	`ssh-keygen -R <ip>`, or `host_key_checking = False` in `ansible.cfg` for labs
`UNREACHABLE ... timed out`	Wrong IP, or the VM is stopped	`multipass list`, then `multipass start ansiblemain`

Files: `section02-getting-started/01-hosts-file/` in the lab repo.